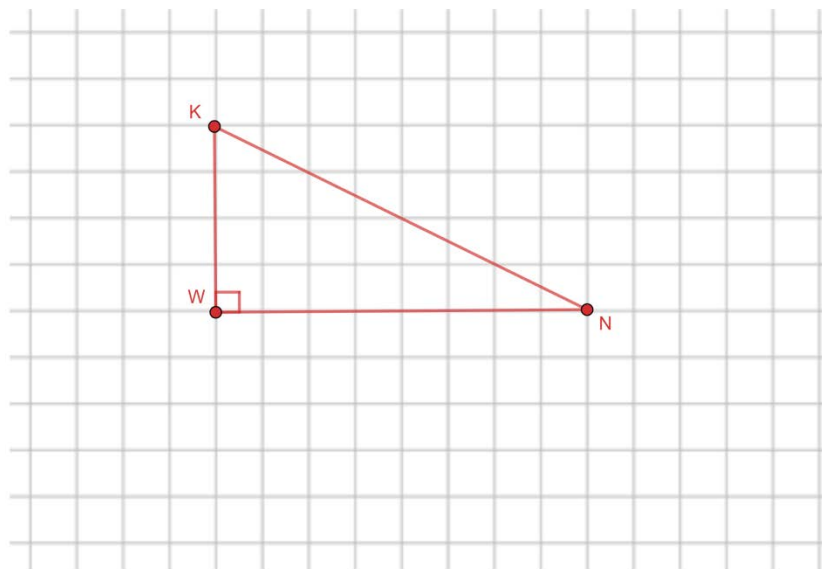
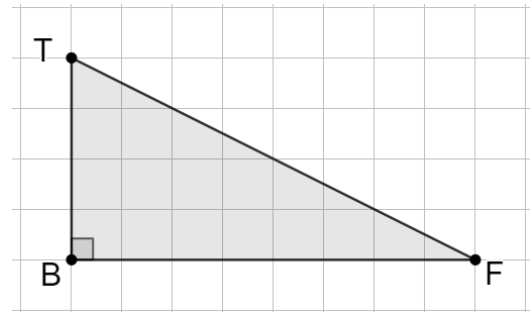


Triangle TBF is shown on a grid.

1. Draw $\overline{KW}$ on the grid so that $KW = BT$.
2. Draw $\overline{WN}$ on the grid so that $WN = BF$ and $\overline{WN}$ is perpendicular to $\overline{KW}$.
3. Draw $\overline{KN}$ using a straightedge.



4. Complete the statements.

By the definition of congruence, since $KW = BT$ then $\overline{KW} \cong \overline{BT}$ and since $WN = BF$ then $\overline{WN} \cong \overline{BF}$. Since $m\angle TBF = 90^\circ$ and $m\angle KWN = 90^\circ$, then by applying the transitive property and the definition of congruence, then $\angle TBF \cong \angle KWN$. The construction of $\triangle KWN$ uses

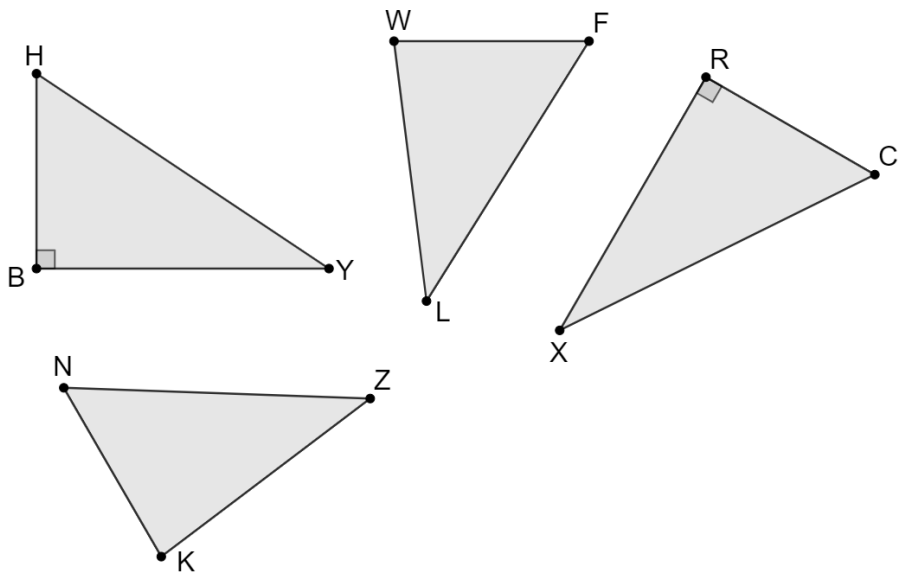
- ☐ ASA
- ☒ SAS
- ☐ SSS

to show that $\triangle TBF \cong \triangle \underline{KWN}$.

Because both triangles are right triangles the triangles could also be shown as congruent by showing that $\overline{KW} \cong \overline{BT}$ and that

$\overline{KN}$ $\cong$ $\overline{TF}$ or that $\overline{WN} \cong \overline{BF}$ and that $\overline{KN}$ $\cong$ $\overline{TF}$.

Triangles KZN , BYH , LWF , and XRC are shown.



Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
5.	$\overline{BY} \cong \overline{XR}$, $\overline{YH} \cong \overline{CX}$	Yes	HL
6.	$\angle NZK \cong \angle WLF$, $\overline{NK} \cong \overline{FW}$, $\angle ZKN \cong \angle FWL$	Yes	AAS
7.	$\overline{HB} \cong \overline{RC}$, $\overline{BY} \cong \overline{XR}$, $\overline{HY} \cong \overline{CX}$	Yes	SSS
8.	$\angle WFL \cong \angle ZNK$, $\overline{WF} \cong \overline{NK}$, $\angle NZK \cong \angle WLF$	Yes	AAS
9.	$\angle ZKN \cong \angle FWL$, $\overline{NZ} \cong \overline{FL}$, $\overline{KZ} \cong \overline{LW}$	No	
10.	$\angle YBH \cong \angle CRX$, $\angle BHY \cong \angle XCR$, $\overline{HY} \cong \overline{CX}$	Yes	AAS